Vel Tech Group of Institutions

Name: Gontu Yaswanth
Email: vtu33434@veltech.edu.in
Roll no: Vtu33434
Phone: null
Branch: VTU
Department: CSE
Batch: 2028
Degree: B.Tech - Computer Science and Engineering

Scan to verify results

2028_NeoColab_Competitive Coding II_L8

VTU_Week 11_MCQ

Attempt : 1
Total Mark : 15
Marks Obtained : 12

Section 1 : MCQ

1. Consider the following two sequences:

X = < B, C, D, C, A, B, C >Y = < C, A, D, B, C, B >

The length of longest common subsequence of X and Y is:

Answer

4

Status : Correct ***Marks : 1/1***

2. Which of the following problems should be solved using dynamic programming?

Answer

Longest common subsequence

Status : Correct **Marks : 1/1**

3. Which of the following problems is closely related to the Subsets Sum problem and can also be solved using dynamic programming?

Answer

Coin Change Problem

Status : Correct **Marks : 1/1**

4. Consider two strings A = "qpqrr" and B = "pqprqrp". Let x be the length of the longest common subsequence (not necessarily contiguous) between A and B and let y be the number of such longest common subsequences between A and B.

Then x + 10y = ______.

Answer

33

Status : Wrong **Marks : 0/1**

5. Consider the strings "PQRSTPQRS" and "PRATPBRQRPS". What is the length of the longest common subsequence?

Answer

7

Status : Correct **Marks : 1/1**

6. Which of the following is/are property/properties of a dynamic programming problem?

Answer

Both optimal substructure and overlapping subproblems

Status : Correct **Marks : 1/1**

7. If an optimal solution can be created for a problem by constructing optimal solutions for its subproblems, the problem possesses _________ property.

Answer

Optimal substructure

Status : Correct ***Marks : 1/1***

8. In dynamic programming, the technique of storing the previously calculated values is called ___________.

Answer

Memoization

Status : Correct ***Marks : 1/1***

9. Which of the following is an example of a problem that can be solved using memoization in dynamic programming?

Answer

Computing the edit distance between two strings

Status : Correct ***Marks : 1/1***

10. When a top-down approach to dynamic programming is applied to a problem, it usually _____________.

Answer

decreases both, the time complexity and the space complexity

Status : Wrong ***Marks : 0/1***

11. Which of the following problems is NOT solved using dynamic programming?

Answer

Fractional knapsack problem

Status : Correct **_Marks : 1/1_**

12. Which of the following is NOT a characteristic of dynamic programming?

Answer

Solving problems in a sequential manner.

Status : Correct **_Marks : 1/1_**

13. What happens when a top-down approach of dynamic programming is applied to any problem?

Answer

It increases the space complexity and decreases the time complexity.

Status : Correct **_Marks : 1/1_**

14. The Fibonacci sequence is often used to illustrate dynamic programming concepts. What is the time complexity of a naive recursive implementation of Fibonacci numbers?

Answer

O(log n)

Status : Wrong **_Marks : 0/1_**

15. What are the different techniques to solve dynamic programming problems:

Answer

Memoization and Bottom-Up

Status : Correct **_Marks : 1/1_**